| Feature | SQL Database | NoSQL Database |
|---|---|---|
| Data structure | Relational tables | various data models(document, key-value, wide-column, graph, etc |
| Schema | Strict schema | Dynamic or schema-less |
| Query Language | SQL(Structured Query Language) | Query language varies by database type (e. g , MongoDB uses BSON queries |
| Scalability | Vertical scaling ( increasing hardware power) | Horizontal scaling (adding more machines to a network) |
| Scaling Type | Mostly scale-up | Mostly scale-out |
| Transactions | ACID (Atomicity, Consistency, Isolation, Durability, Consistency Isolation, Durability)properties, | BASE(Basically Available, Soft state, Eventually consistent) properties |
| Use cases | Well-suited for complex queries and transactions. | Well-suited for large-scale, distributed, and rapidly changing data |

| | | |
|---|---|---|
| Example Database | MySQL, PostgreSQL, Oracle | MongoDB, Cassandra, Couchbase |
| Data Relationship | Emphasizes, relationships through foreign keys | Relationships handled through embedding referencing, or graph-based approaches |
| Flexibility | Rigid schema structure, changes may be complex | Flexible schema, easy to add or modify fields |
| Consistency Model | Strong consistency | Eventual consistency (varies by implementation) |
| Complexity | Well-established and extensive community support | Generally easier to scale and manage, especially in distributed environments |
| Community Support | Typically more complex to set up and manage | Growing community support, specific to each NoSQL technology |
| Example | MySQL, PostgreSQL, SQL Server, Oracle | MongoDB, Cassandra, CouchDB, Redis |

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